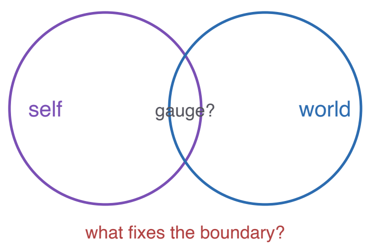
What It All Means

A Philosophy Primer on Concern, Selfhood, and Meaning in an Empirical Research Program



A companion to *The Mathematics of Constraint*.

The program uses words — concern, valence, self, care, meaning — that carry centuries of philosophy. This book asks what it is entitled to mean by them.

Why Philosophy, and Why Carefully

This research program runs on words that are not innocent. *Concern. Valence. Self. Care. Meaning. Agency.* Each carries a heavy freight of philosophy, and each invites the reader to hear more than the experiments have shown. When a paper says an agent "maintains its self/world boundary," you can hear a claim about minds — or a description of a small simulated creature deciding whether to eat a pixel. Both can be true at once, and the entire philosophical task is keeping straight which one is being asserted.

This book does two things a mature reader needs. It takes the program's philosophical claims seriously — because several are genuinely interesting, and one or two may be real contributions to the philosophy of mind. And it holds them to account — because a program that borrows the vocabulary of care and selfhood owes an argument that it has *earned* the words, not merely relabeled reward signals with honorifics. The goal is that you finish able to see exactly where the program is doing real philosophy, where it is doing suggestive metaphor, and where a sharp critic should press.

A note on fairness

The program is, to its great credit, *deliberately deflationary about its own reach*. Its repeated refusals — "we do not claim consciousness" — are not throat-clearing; they are load-bearing to its credibility, and this book treats them as seriously as its claims. The Thesis boxes state what the program asserts; the Tension boxes press where a philosopher should; the Intuition pump boxes supply the picture. The aim is neither to debunk nor to celebrate, but to make you a fair and demanding reader.

Contents

PART I

The Central Theses

The program has a philosophical spine — a small set of theses about meaning, generalization, and objecthood that its experiments are built to test. This part states each one plainly, in the program's own terms, and gives the picture that makes it intelligible. Part II then turns to the hard problems they run into.

Claim map — what kind of statement is each thesis?

This primer uses six claim tiers: a *definition* fixes vocabulary; a *motivational frame* organizes questions; a *formal result* follows under stated assumptions; a *controlled empirical result* is bounded by its tested substrate; an *extrapolation* is a registered research bet; and a *disclaimed claim* is not supported by this program.

Topic	Tier and evidence	Scope	Falsifier or blocking result
Meaning	Motivational frame; concern-geometry and intervention experiments	Toy and controlled learned representations	Geometry tracks supplied labels but neither self-observed stakes nor decisions under intervention
Concern	Definition C1/C2 below; controlled C2 evidence in staged ΔE learning	Supplied weights or self-observed viability signals, never felt mattering	Yoked/wrong-value signals perform equally, or the signal is available but not causally used
Weakness	Formal conditional result plus bounded internal evidence	Declared transformation family and matched deployment regime	Group/prior misspecification reverses selection; the external Pythia kill already blocks universality
Objects	Controlled empirical result	Supervised causal-role labels in toy environments	Appearance-matched or wrong-role controls yield the same stable/useful categories
Self	Formal non-identifiability and model-relative intervention result	A stated SCM and allowed null-action interventions	Two admissible gauges make different held-out counterfactual predictions without a deciding intervention
Use	Definition plus controlled causal-intervention result	A specified feature, intervention family, commitment surface, and transport set	Matched wrong-subspace or off-manifold interventions explain the same dose-response

Agency	Extrapolation from homeostasis, causal use, robustness, and supervised repair	Agent-like control; strict autopoiesis remains untested	No autonomous return dynamics, endogenous regulation, or component/boundary production
Consciousness	Disclaimed claim	At most structural correlates in systems independently known to be conscious	No experiment here can certify phenomenal experience or moral status

CHAPTER 1

Meaning Is Geometry Under Concern

1.1 The founding axiom

At the base of the program sits a single sentence, repeated in various forms across its documents: **meaning is geometry under concern.** Unpacked into its steps, the claim is a chain: nothing appears without *difference*; organized difference is *geometry*; geometry becomes *meaning* when some differences are *salient* — when they matter to a system that has stakes; and a system has stakes when it has *concern*, a caring about its own continuation. So meaning, on this view, is not a property of symbols or a relation of resemblance. It is what structured difference *becomes* for a system that can be helped or harmed.

Intuition pump — the map that only means something to a traveler

A contour map is just ink until someone needs to cross the terrain. To a traveler on foot, the tight contour lines *mean* "exhausting climb"; to a helicopter pilot they mean almost nothing. The marks are identical; the meaning is supplied by whose stakes are in play. The program's axiom generalizes this: the geometry of a representation is inert structure until a system with something to lose reads it under concern. "Meaning is geometry under concern" says the meaning was never in the geometry alone — it was in the meeting of geometry with a stake. This is why the program insists that to study meaning you must study systems that can *die* (or fail, or be harmed), because only there does concern, and therefore meaning, get a grip.

1.2 The lineage the axiom compresses

This is not a new idea so much as a *synthesis* of an old and honorable tradition. It descends from **phenomenology** (Heidegger's *care* as the basic structure of a being for whom its own existence is at issue), from **ecological psychology** (Gibson's affordances — the world shows up in terms of what it offers *for action*), from **Uexküll's Umwelt** (each creature inhabits a world carved by what matters to it), and from **enactivism** (meaning as sense-making by a self-maintaining organism). The program's own honest positioning is that these traditions "predicted the shape" — that meaning is care-laden, action-oriented, embodied, boundary-maintaining — while the program supplies the *mechanism* and the *measurement*. It is philosophy's intuition rendered as an experiment.

1.3 The two-part structure: meaning, then agency

The axiom has a second half that is easy to miss and does most of the program's work. *Meaning* appears when differences become salient under concern; *agency* appears when that concern is "coupled to action, self-maintenance, boundary preservation, repair, and time." So the program draws a line between two grades of structure. A system can have *meaning-like* structure — a geometry organized by what matters — without yet being an *agent*. It becomes agent-like only when the concern-organized geometry starts to *act to preserve itself*. This distinction — passive meaning-geometry versus active self-maintaining agency — is the hinge of the entire program, and it recurs in every part of this book.

Intuition pump — the thermostat that started defending itself

A thermostat has a kind of minimal "concern" — a setpoint it works to maintain. But it is not an agent in any robust sense; it is a passive loop. Now imagine a thermostat that, when you tried to break it, rewired itself to keep regulating; that treated its own continued functioning as something to protect; that, over time, reorganized its sensors around what threatened it. Somewhere along that escalation, "a device with a setpoint" starts to feel like "a thing that cares about persisting." The program's central bet is that this escalation is *real and measurable* — that there is a threshold where passive concern-geometry becomes active self-maintenance — and that finding and characterizing that threshold is the deep prize. Whether the threshold is a genuine transition or a gradual metaphor is a question Part II and the complexity primer both press.

1.4 Why "under concern" is the load-bearing phrase

Strip "concern" out of the axiom and you are left with "meaning is geometry" — which is just the observation that representations have structure, true but empty. The whole content is in "under concern": the claim that *what a system cares about* is what turns its structure into meaning. This is what makes the axiom a *substantive* and *testable* claim rather than a definition. If it is right, then changing what a system cares about should reorganize its geometry — which is exactly the experiment Chapter 3 describes. And if concern turns out to be something the experimenter simply *supplies* rather than something the system *has*, the axiom risks collapsing into "geometry follows whatever we weight" — the central tension of Chapter 4. So the axiom's whole fate rides on one word, and Part II is largely the interrogation of that word.

1.5 What to carry forward

The founding axiom is "meaning is geometry under concern" — structured difference becomes meaning for a system with stakes; it synthesizes phenomenology, affordances, Umwelt, and enactivism into a mechanism-and-measurement program; it distinguishes passive meaning-geometry from active self-maintaining agency (the program's central hinge); and everything rides on the word "concern," whose status — genuinely the system's, or merely the experimenter's — is the tension the rest of the book pursues.

CHAPTER 2

Weakness and the Metaphysics of Generalization

2.1 A philosophical claim disguised as a machine-learning result

The program's most famous result — that "weakness" predicts generalization better than simplicity — looks like a technical finding in machine learning. It is also a philosophical thesis about **induction**: about what makes one extrapolation from evidence better than another. That problem — Hume's problem, the problem of why any inference from the observed to the unobserved is justified — is among the oldest in philosophy, and the program takes a definite side in it.

Thesis — the weakest hypothesis, not the shortest

Faced with many rules that fit the evidence equally well, the classical answer (Occam, and its precise modern form, Minimum Description Length) is: prefer the *simplest* — the shortest description. The program, following Bennett, asserts a rival: prefer the *weakest* — the hypothesis compatible with the most possibilities while still fitting. The slogan, inherited from Bennett: *simplicity is a property of form, but generalization is a property of function*. A short description can be brittle; a weak hypothesis, having committed to less, travels further when the world shifts.

Intuition pump — the overconfident witness

Two witnesses describe a suspect. The first gives an extremely specific account: exactly this height, this jacket, this gait, this scar. The second says only: "left-handed." The specific account is *simpler* in one sense — it is a single crisp hypothesis — and it fits the evidence seen so far perfectly. But when a new detail emerges that the first witness couldn't have known, their over-committed story is often simply *wrong*, while the second witness's minimal, broadly- compatible claim survives. The program's wager is that under distribution shift — when deployment differs from training — the weak, under-committed hypothesis generalizes because it never bet the farm on the training set's accidents. Weakness is disciplined humility about what you have not yet seen. This is a genuine philosophical position on induction, not merely an engineering trick.

2.2 What the program adds to the philosophy

Bennett's claim that weakness beats simplicity was, before this program, an argument. The program's contribution is to make it an *experiment* — and this is philosophically significant in its own right. It constructs situations where a simple shortcut and a weak symmetric rule both fit the training data, then measures which one generalizes. The result (weakness wins near-universally; every simplicity measure loses) does something philosophy rarely can: it turns a dispute about the nature of good inference into a decidable question with an answer. That a metaphysical thesis about induction can be operationalized and tested at all is part of what makes the program interesting to a philosopher, independent of whether one accepts the specific result.

2.3 The reparameterization worry, and why weakness answers it

A deep objection to simplicity-based induction is that "shortest" depends on your *language*: a rule that is short in one coding scheme is long in another, so "prefer the shortest" secretly smuggles in a choice of description language. Weakness has a partial answer: it is defined in terms of *which possibilities a hypothesis permits* — a property of the function's behavior, not its notation — and so is invariant under relabeling. This is the substantive content of "generalization is a property of function, not form": weakness sidesteps the language-dependence that has always haunted Occam's razor. Whether it fully escapes it (you still must choose which *symmetry group* to count against) is a real question, and one the program itself surfaces.

Tension — is the weakness result a discovery or a near-tautology?

A mature reader should hold two deflationary worries the program half-concedes. First, the bridge theorems connecting weakness, symmetry, and concern are, in the program's own words, "almost tautological" — they show where concern *enters* the bookkeeping rather than deriving a surprising fact. The empirical bite is entirely in the measured correlation, not the proofs. Second, and more seriously: the external "hard-kill" (weakness failing to predict generalization on a model the lab didn't build) suggests the result may be bound to the specific regime the program constructs — tasks where a symmetry genuinely generates the deployment data. Weakness may be less a universal law of induction than a correct principle *within a class of problems the program is good at building*. The philosophical thesis (weakness over simplicity) is genuine and interesting; its scope is narrower and more contested than the headline suggests, and the program's own reframe — "weakness is a footprint, compatibility is the cause" — is the admission of exactly this.

2.4 The metaphysical upshot

Step back from the machinery and the thesis is this: *good generalization is under-commitment to what you have seen, in favor of compatibility with a structure that generates what you haven't*. It reframes induction from a search for the simplest story to a search for the story that keeps the most futures open while still fitting the past. This is a substantive and attractive position in the philosophy of induction — arguably a sharpening of the intuition behind "robustness" and "invariance" that a good scientific hypothesis should have. The program's service to philosophy here is less the specific correlation than the demonstration that this ancient dispute can be dragged onto experimental ground at all.

2.5 What to carry forward

The weakness result is a philosophical thesis about induction: prefer the weakest (most broadly-compatible) hypothesis, not the shortest, because "generalization is a property of function, not form"; the program's contribution is turning a metaphysical dispute into a decidable experiment; weakness partly escapes the language-dependence that haunts Occam; but the bridge theorems are near-tautological and the external hard-kill suggests the result's scope is narrower than the headline — the thesis is genuine, its universality contested.

CHAPTER 3

Objects Form From Concern

3.1 A claim about natural kinds

The program's third central thesis is the most metaphysically ambitious: that a system carves the world into *objects* — natural kinds, categories, "things" — not by **sensory similarity** but by **causal-valence role**: by what matters for its viability. Two items that look completely different but affect the agent's survival the same way get grouped together; two that look identical but have opposite consequences get split apart. The world's joints, on this view, are carved by concern, not by appearance.

Thesis — carving by role, not resemblance

"Meaning is not sensory resemblance; meaning is action-relevant, valenced causal structure. Objects can form from concern." The program's cleanest experiment: three identical networks are trained with different objectives on the same inputs. The one whose objective is coupled to the agent's viability organizes its internal space almost entirely around the *reward axis* (the thing that determines the right action) and *discards* the sensory features — even under a task (XOR) where neither sensory feature alone predicts reward. The measured geometry is stark: the reward-axis organization is hundreds of times stronger than the sensory-axis organization.

Intuition pump — the forager's ontology

A forager does not sort the forest by color or leaf shape; she sorts it into *food, medicine, poison, useless* — categories defined entirely by what each plant does to a body that eats it. Two unrelated-looking berries that both nourish are, to her ontology, "the same kind of thing"; a nourishing berry and its poisonous look-alike are opposites, however similar to the eye. Her world is carved by consequence, not appearance. The program's claim is that *any* viability-coupled learner carves this way — that concern is the knife that finds the joints. It is a striking and, in the toy setting, genuinely measured phenomenon: give a network stakes, and its categories reorganize around the stakes.

3.2 The philosophical stakes: a theory of content

This is, whether or not the program names it as such, a contribution to the philosophy of **mental content** — the question of what makes an internal state be *about* something. The dominant naturalistic answer, **teleosemantics** (Millikan, Dretske, Papineau), holds that a state's content is grounded in its *function* — what it was selected or learned *to do* for the organism. "Objects form from concern" is a computational teleosemantics: the content of the network's categories is grounded in their function for viability. This connection is the program's most promising unexploited opening — because teleosemantics comes with forty years of sharpened objections and defenses (the Swampman problem, the indeterminacy of function) that would let the program state its claim precisely and defend it against known attacks. That the program reaches this territory without naming it is a scholarly gap; that it reaches it at all is a real contribution.

3.3 The natural-kinds connection

"Carving by causal role" is also a position on the ancient question of **natural kinds** — whether the world's categories are *discovered* (real joints, out there) or *constituted* (useful groupings we impose). The most sophisticated modern view, Boyd's **homeostatic property cluster** theory, holds that natural kinds are exactly clusters of properties held together by an underlying causal mechanism — which is remarkably close to "objects are clusters held together by shared causal-valence role." The program's kinds are neither purely discovered nor purely imposed: they are *real relative to a system's viability* — the joints a creature-with-stakes must find to survive. This is a defensible and interesting middle position, and naming Boyd would let the program claim it precisely.

Tension — is the carving discovered, or installed by supervision?

Here is the sharpest philosophical worry, and it is the same one that stalks the whole program (Chapter 4). In the cleanest object-formation experiment, the "valence-coupled" network is trained to predict the *optimal action* under the reward function — which is a *supervised* signal. So a deflationary reading is available: the network reorganizes around the reward axis because the experimenter *told it to* predict a reward-derived label. On this reading, "objects form from concern" is really "objects form from supervised labels of concern" — which is far less surprising, and much weaker as a metaphysics. The first joint sparse-reward REINFORCE attempt did fail on XOR, but that is no longer the final result: staged learning from observed energy changes reaches a reward-geometry gap of +1.84 without optimal-action labels, and a ΔE-argmax planner reaches return 50/50. Those bounded toy results close the old "XOR cannot work" claim, while leaving a stronger question open: the pipeline still relies on uniform exploration, observed ΔE, and staged modules rather than demonstrating robust, endogenous concern under ecological shift and adversarial signal controls.

3.4 What is genuinely shown, and what is aspired to

To be fair to both sides: the program has genuinely shown, with clean controls, that *if* a learner's objective is coupled to viability, its representational geometry reorganizes around causal role and discards sensory similarity — a real, measured dissociation between appearance-based and consequence-based representation. That is a solid result and a nice illustration of a teleosemantic mechanism. Later staged work also learns useful reward geometry and action from self-observed ΔE without optimal-action labels. What it has *not* yet shown is the stronger claim the vocabulary implies: that this is the system's own concern rather than any correlated scalar the experimenter makes available. Yoked-agent, sign-shuffled, wrong-variable, and shifted-dynamics controls are still needed to separate endogenous stakes from successful value-signal engineering.

3.5 What to carry forward

The thesis "objects form from concern" claims the world is carved into kinds by causal-valence role, not sensory resemblance — a measured dissociation in the repo; philosophically it is a computational teleosemantics (content grounded in function-for-viability) and a Boyd-style homeostatic-cluster theory of natural kinds, both unnamed but reached; and the standing tension is that the cleanest demonstration uses *supervised* concern-labels, so the strong "a system carves its own world from its own stakes" claim is only partly earned.

PART II

The Hard Problems

The theses of Part I run, on contact with hard cases, into four deep problems: what concern really is, whether the self is even determinate, when a representation is genuinely used, and whether any of this touches consciousness. These are not weaknesses to be embarrassed by; they are where the program is doing its most interesting philosophical work, and where a demanding reader should engage most closely.

CHAPTER 4

What Is Concern, Really?

4.1 The word doing the most work, defined the most thinly

"Concern" is the keystone of the whole program — meaning is geometry *under concern*, objects form *from concern* — and it is also the term whose definition is most contested. The program is admirably explicit that it does *not* mean concern in the felt, phenomenal sense. It means something operational: *a nonnegative weighting over the distinctions that matter for a system's future control, viability, action, or evaluation*. In the experiments, this cashes out concretely as a loss weight, or as the predicted change in the agent's energy, or as the sign of a reward. Concern is, operationally, a number the system assigns to outcomes.

The program's own guard

The program states the deflation itself, in several papers: "The phrase 'concern' here is not used as a folk-psychological attitude. It is a nonnegative measure over distinctions that matter for the system's future control." And, bluntly, about the gap between the operational and the felt: "Loss-weight ≠ hunger/wound/life. No consciousness, subjectivity, biological realism." The program knows the word is carrying more than the math, and says so.

4.2 The imported-versus-derived problem

Here is the deepest philosophical tension in the entire program, and it is worth stating with care because everything downstream depends on it. If concern is *defined* as a weighting the experimenter supplies, then "meaning is geometry under concern" is in danger of reducing to "geometry follows whatever we weight." The *mattering* — the thing that was supposed to make geometry into meaning — would be **imported** by the researcher, not **derived** from the system. A network reorganizing around a reward axis *because we trained it to predict reward* is not obviously a system whose *own* concern shapes its *own* meaning; it is a system doing what its loss function selects.

Intuition pump — the puppet's values

A puppeteer can make a puppet "flinch from fire" and "reach for food." Watching, you might say the puppet cares about survival. But the caring is entirely the puppeteer's — pull the strings differently and the puppet "cares" about the opposite. Nothing in the puppet *has* the value; the value is imposed from outside. The worry about concern is precisely this: when the experimenter sets the reward and trains the network to track it, is the resulting "concern-geometry" the network's, or the experimenter's, ventriloquized through it? The program's grand thesis needs concern to be the puppet's, not the puppeteer's — and the honest state of the evidence is that, in most experiments, the strings are visible.

4.3 The bootstrap attempt, and its partial failure

The program sees this problem clearly and tries to solve it. Its "concern-bootstrap" experiments attempt to make concern **self-organize** — to have the agent learn what matters from its *own* observed experience (the energy changes it actually undergoes) rather than from a supervised optimal-action label. The initial joint REINFORCE learner failed on nonlinear XOR. The later diagnosis split representation from competence: staged ΔE learning reached a +1.84 reward-geometry gap and ΔE-argmax planning reached 50/50. This is real progress from interaction-derived signals, but uniform exploration, directly observed ΔE, staged training, and toy scale keep it below a general claim that the system's own concern spontaneously shapes meaning.

4.4 The three senses of concern, and why disentangling them would help

A constructive diagnosis: the program conflates three genuinely different things under one word, and its arguments would sharpen enormously if it separated them:

Sense	What it is	Status
C1 — Supplied weight	a loss weighting the experimenter chooses	clearly in play; unproblematic but philosophically thin
C2 — Self-observed viability signal	the agent's own registered change in its energy/damage — a value it computes from its own state	the interesting case; partly demonstrated by the staged ΔE work
C3 — Phenomenal mattering	felt care, the "what it is like" to have stakes	explicitly out of scope; the program disclaims it

Canonical usage: every empirical use of “concern” in this program must be read as **C1** or **C2**. C1 includes explicit loss weights, reward signs, supplied costs, and supervised valence labels. C2 requires a signal computed from the system's own observed state transition and must beat yoked, wrong-value, and shuffled-signal controls. C3 is disowned. The whole force of “meaning is geometry under concern” therefore depends on moving evidence from C1 toward C2 without silently borrowing C3.

Experiment family	Concern sense actually tested	Boundary
Concern-weighted weakness; supervised object formation	C1	Shows what supplied stakes organize, not whose stakes they are
E1/E4 commitment-surface interventions	C1 with causal-use tests	Load-bearing supplied concern; E5 shows labeled coverage can mimic rule learning
Joint XOR bootstrap	Attempted C2; failed	Failure remains part of the correction chain
Staged ΔE geometry and ΔE planning	Bounded C2	+1.84 geometry gap and 50/50 planning under uniform exploration and directly observed ΔE

Any consciousness-adjacent interpretation	Not C3 evidence	No phenomenal or moral-status inference

Tension — the honest one-line objection

A hard-nosed philosopher's summary, which the program must answer: *"You showed that if you weight a loss by X, representations reorganize around X. That is expected, not a discovery about meaning."* The program's best reply is not to deny it but to point at sense (ii) and the load-bearing standard of Chapter 6: the claim is not merely that weighting reorganizes geometry, but that only geometry which *survives to causally control the decision under intervention* counts as meaningful — and that this criterion is doing real work, filtering honorific relabeling from earned attribution. Whether that reply fully rescues the grand thesis is genuinely open. But the reply exists, and it is where the program's philosophical action really lives.

4.5 What to carry forward

Concern is the keystone term, defined operationally as a nonnegative weighting (a loss weight, an energy change, a reward sign), with felt concern explicitly disclaimed; the deep tension is whether concern is imported by the experimenter or derived from the system — "geometry follows whatever we weight" versus a system's own stakes; the bootstrap experiments try to derive it and partly succeed; and the fix is to disentangle supplied-weight, self-observed-viability, and phenomenal senses, committing publicly to the first two and building toward the second.

CHAPTER 5

The Self/World Gauge Problem

5.1 The program's most genuinely novel philosophical result

Of all the program's philosophical contributions, this is the one a philosopher of mind should take most seriously, because it is a *proof* — a clean, formal result — about the determinacy of the self. The claim: **a system can be behaviorally a perfect agent while having no fact of the matter about which changes in the world are "its own."** The self, on this result, is not something a system reads off from its experience; it is structurally under-determined by any amount of passive observation, and can only be fixed by intervention.

5.2 The setup, in plain terms

Give an agent the task of splitting each change it observes into two parts: the part *it* caused (the "self" contribution) and the part the *world* caused (the "world" contribution). The architecture is built to do exactly this — two sub-networks, one for self, one for world, summing to predict the total. Here is the problem: the environment only ever reveals the *total* change. And you can add any constant to the "self" estimate and subtract it from the "world" estimate without changing the total by one iota — so the training signal, which only sees the total, is completely *blind* to where the boundary falls. The network settles on *some* split, but which one is determined by its random initialization, not by any fact about what it actually caused.

Definition — a gauge symmetry (borrowed from physics)

In physics, a **gauge symmetry** is a change in your *description* that leaves every observable quantity unchanged — like adding a constant to all altitudes: no dropped ball falls differently, because only altitude *differences* are real. The choice of "zero altitude" is a gauge: pure convention, carrying no physical content. The program's result is that the self/world split is exactly such a gauge. The "self" and "world" labels are like the choice of zero — you can slide the boundary freely without changing anything the system does. The program states the resulting no-go theorem crisply: if observations are invariant under a gauge, no procedure using only those observations can identify which description is the "true" one.

Intuition pump — splitting a restaurant bill you never itemized

Two friends share a $60 dinner and want to split it into "what I ordered" and "what you ordered" — but the receipt shows only the $60 total. Any split that sums to $60 is consistent with the receipt: $40/$20, $30/$30, $50/$10. The data simply does not contain the answer. No amount of *staring* at the receipt will fix it; the information isn't there. The only way to learn who ordered what is to *intervene* — to have one friend order nothing next time and see what the bill becomes. The self/world boundary is this bill. The program's insight is that a network asked to itemize a total it can only observe as a sum is in a genuinely hopeless position — and that the escape is not more observation but *action*: the agent deliberately doing nothing, to see the world's contribution alone.

5.3 Why this is a real philosophy-of-mind result

The philosophical payload is substantial. A long tradition — running from Kant through Hume to modern predictive-processing theorists like Metzinger — has argued that the self is not *given* in experience but is a structural *posit*, a construction the mind imposes rather than a datum it finds. The program's gauge result is a **precise, computational vindication of this ancient worry**. It shows, in a concrete trainable system, that "which part of this is me?" is not answerable from the inside by observation alone — that the self is a gauge choice, fixable only by intervening on the world. That a philosophical intuition about the constructed, non-given nature of the self can be turned into a formal identifiability theorem in a neural network is genuinely new, and it is the program's strongest claim to have contributed something to philosophy rather than merely borrowed from it.

5.4 The resolution: the self is known by not acting

The program's fix is as philosophically resonant as the problem. The gauge is broken by the *null action* — the agent deliberately doing nothing — which lets it observe the world's contribution in isolation and thereby pin down the previously-free boundary. This connects directly to the neuroscience of **reafference** (von Holst, Sperry): the brain distinguishes self-caused from world-caused sensation by predicting the consequences of its own actions ("corollary discharge"). The program's result generalizes this into a principle: *the self is learned by intervention, not observation — the agent learns its own boundary by deliberately not acting*. There is something philosophically satisfying in the conclusion that selfhood is grounded in agency (in what one does and refrains from doing) rather than in introspection.

Tension — does intervention recover a real self, or just a chosen frame?

One genuine unresolved tension a philosopher should press. The program says two things that are in mild tension. On one hand: the architectural prior that there *is* a self/world split is "a frame, not a fact" — the boundary is under-determined, a construction. On the other: the null-action intervention "recovers the true decomposition." The defensible repo-grounded claim is narrower: null anchoring identifies a chosen operational decomposition relative to the declared structural model and allowed intervention set; richer shared-head worlds remain under-identified. What, then, makes that intervention-fixed version *true* rather than just another practically-useful choice of frame? Is the self that intervention reveals a *real* self that was there all along, or a *useful* self that intervention *constructs*? This is the deep question — realism versus instrumentalism about the self — and the program's result sharpens it without settling it. That is not a flaw; sharpening a genuine question to the point where it becomes decidable is exactly what good philosophy of mind does, and this result does it.

5.5 What to carry forward

The gauge result proves a system can be a perfect agent with no fact of the matter about which changes are its own — the self/world split is a gauge symmetry, under-determined by observation, a no-go theorem for passive identification; it is a computational vindication of the Kant-to-Metzinger view that the self is posited, not given; null intervention fixes an operational gauge within the stated causal model rather than recovering a metaphysically unique self; and it leaves open a sharpened realism-versus-instrumentalism question about whether intervention reveals or constructs the self.

CHAPTER 6

Availability Is Not Use

6.1 The distinction that may be the program's most durable idea

Running through the whole program is a distinction so important that it survives every one of the program's own negative results — indeed, it *produces* them. The distinction is between a representation being **available** and being **used**. That a concept can be *decoded* from a system's internal states — that a probe can read it off with high accuracy — does *not* establish that the system *uses* the concept to make its decisions. Availability is cheap and common; use is what matters, and it must be shown separately.

Thesis — the load-bearing standard

A representation is "load-bearing" for a decision only if intervening on it — surgically corrupting or removing it — actually changes the decision at the point where the decision is made (the "commitment surface"). The program even proves an impossibility result: two features can have *identical* decodability (identical probe accuracy) while their *causal* effects on the decision differ arbitrarily. Therefore decodability can never, in principle, establish use. Only intervention can.

Intuition pump — the books on the shelf you never read

Your bookshelf makes a great deal of information "available" — every fact in every volume is right there, decodable by anyone who opens a book. But which of it do *you* actually *use* to make decisions? Only the small fraction you have read and internalized. An observer cataloguing your shelf would badly overestimate your knowledge if they assumed availability equals use. Now run the test properly: *remove* a book and see if your behavior changes. If it does, that book was load-bearing; if not, it was decoration. The program insists on running exactly this test on a neural network's internal features — not "can we read it off?" (the shelf) but "does removing it change the decision?" (the reading). A probe reading a feature is finding a book on the shelf; only patching it proves the system ever read it.

6.2 Why this matters beyond the program

This is a genuine contribution to the philosophy of mind and to the young field of AI interpretability, and it is worth appreciating why. A great deal of interpretability research implicitly assumes that if a concept is decodable from a model's activations, the model "represents and uses" it. The program's result says this inference is *invalid* — decodability and use are logically independent, provably so. This reconstructs, in a modern computational setting, a distinction the philosopher Fred Dretske drew decades ago between a **structuring cause** (what explains why a triggering event is wired to produce the behavior) and a **triggering cause** (the event that occasions the behavior). Dretske's history-sensitive distinction now constrains, rather than merely resembles, the experiment: an arbitrary activation patch is not automatically a structuring cause. Dose-response evidence must be paired with learning-history, mechanism-substitution, and transport controls before the stronger analogy is earned.

6.3 The other edge: success without representation

The distinction cuts both ways, and the program is honest about the uncomfortable direction. Just as a representation can be available without being used, a system can *succeed* at a task without representing the thing you thought it needed to represent. In one experiment, an agent solves its viability problem *without representing its viability axis at all* — it has, in the program's words, "acquired a functional disposition, not a representational structure." This is the deflationary mirror of the load-bearing standard: behavioral success is not evidence of the intended internal structure. It is the empirical form of a very old debate — **behaviorism versus cognitivism** — and the program lands, with measurements, on the side that says: getting the right answer tells you nothing, by itself, about what is going on inside.

6.4 The criterion of representational reality

Assembled, availability-is-not-use gives the program what may be its single most defensible and general philosophical thesis: a **criterion of representational reality**. It says: for any attribution of a mental or representational property to a system, you must exhibit the *proxy* that would pass the behavioral test *without* the property, and then show the property is *load-bearing* at the surface where the decision is made. In the program's own words: "Do not ask only whether the agent succeeded; ask whether the future-controlling structure reached the surface where success was chosen." This is a rule for when the intentional, representational vocabulary is *earned* rather than merely projected — and it is exactly the discipline that generates all the program's honest negatives, because it keeps catching cases where the vocabulary was projected.

Tension — a criterion still needs a threshold, and a judge

Two honest limits. First, "load-bearing" is stated as a threshold (the intervention changes the decision by *some* amount), but the philosophically interesting quantity is a *degree*: how *much* of the decision does this structure explain? A binary "used / not used" is a coarse rendering of what is really a graded notion of representational involvement, and the program would deepen by formalizing the degree. Second, like every criterion in an AI-generated program, the judgment of whether a given intervention "counts" as reaching the commitment surface is applied by the same kind of system being judged. The criterion is genuinely general and genuinely valuable; making it graded, and having it applied by an independent hand, is what would turn a sharp idea into a settled tool.

Definition — graded, intervention-relative causal use

Fix a feature *z*, a commitment target *Y*, an admissible intervention family, and a transport set *T*. At removed mass α, define the specific effect **D(α) = E[loss(Y) after ablating z at α] − E[loss(Y) after a matched wrong-subspace ablation at α]**. The graded use score is the normalized area under the positive dose-response, **$U = \int D(\alpha)_{+} d\alpha / Z$**, reported with an uncertainty interval, the α range, the normalizer Z, and every surface in T. U is evidence of use only relative to those choices. A paper may still preregister a binary gate for decision-making, but the underlying involvement remains a curve, not an intrinsic yes/no property. Off-manifold patches, width scaling, activation rescaling, and failed transport block the strongest interpretation.

6.5 What to carry forward

Availability (decodability) is provably independent of use (causal effect at the decision) — two features can have identical probe accuracy and arbitrarily different causal impact, so only intervention establishes use; this reconstructs Dretske's structuring-versus-triggering cause and corrects a common interpretability fallacy; it cuts both ways (systems succeed without the expected representation); and assembled it yields a general "criterion of representational reality" — the program's most durable philosophical thesis — now expressed as a graded dose-response U while retaining preregistered publication gates and the need for an external judge.

CHAPTER 7

The Consciousness Question

7.1 The elephant the vocabulary keeps summoning

A program that talks about concern, valence, selves, and care is a program constantly gesturing toward consciousness, whether it wants to or not. These are the words we use for beings with inner lives. So the reader inevitably asks: is any of this a theory of consciousness? Does a system that "maintains its self/world boundary" and "carves the world by concern" have — or approach having — subjective experience? The program's answer is a firm, systematic, repeated **no**, and how it handles this refusal is one of its most philosophically disciplined features.

7.2 The systematic refusal

Nearly every paper carries a near-verbatim disclaimer: "We study minimal computational precursors of concern-like agency. We do not claim consciousness, full agency, or general intelligence." Selfhood results are labeled "a selfhood ingredient, not a consciousness certificate." A standing list enumerates what is explicitly *not* claimed: consciousness, sentience, subjective experience, moral status, human cognitive validity, neural validity. This is not scattered hedging; it is a policy, applied consistently, and it is load-bearing to the program's credibility precisely *because* the vocabulary invites the opposite reading. A program using the word "concern" while refusing the consciousness inference is doing something harder than either using neutral jargon or making the bold claim — it is holding a suggestive vocabulary and declining to cash the suggestion.

Intuition pump — naming the storm "Concern" without believing it feels

Meteorologists name hurricanes with human names — Katrina, Sandy — and describe them as "angry," "unpredictable," "seeking the coast." No one thinks the storm has feelings; the names are handles for tracking and reasoning, and the anthropomorphism is understood as a convenience. The program's use of "concern" and "self" aspires to the same status: technical handles for real structures (viability weightings, causal-attribution boundaries), borrowed from mentalistic vocabulary because that vocabulary carved the right joints first, but explicitly *not* claiming the inner life the words usually imply. The discipline is in never letting the handle quietly become the claim — and the repeated disclaimers are how the program enforces that on itself.

7.3 The one place consciousness is engaged — and how carefully

There is exactly one thread where the program lets consciousness in, and even there it is fenced. In engaging the "geometry of consciousness" work (the ring and torus attractors of head-direction and grid cells, and how their topology *degrades* across brain states — intact in waking and REM sleep, collapsing in deep non-REM), the program floats a *candidate correlate*: preserved topology plus ordered dynamics might mark conscious-like states, degraded topology their absence. But it flags this explicitly as "a candidate operationalization, not a claim." This is the right posture: a *deflationary structural correlate* — a measurable geometric signature that *accompanies* conscious states in the one system (brains) where we independently know consciousness is present — offered as a research hypothesis, not a theory of why there is something it is like to be anything.

7.4 The tapestry of valence, and an honest negative

The program inherits from Bennett a striking speculative idea: that *qualia* — the felt qualities of experience — might be "tapestries of valence," patterns of how things help or harm across an organism's many internal variables. This is a genuine (if highly speculative) proposal about the nature of felt experience. And here the program does something admirable: it tests the geometric form of the claim and reports that it is **not supported** — the network does not visibly cluster its representations by valence-effect role the way the theory predicts. The program carries the tapestry idea in its *functional* form (agents do track multi-dimensional consequences) but honestly reports that its *geometric* form (the tapestry is visible in the representation) failed. It declines to claim the qualia theory its vocabulary would license, on the strength of its own measurement.

Tension — the hard problem is untouched, and that is fine

A philosopher should be clear about what none of this addresses: the **hard problem of consciousness** — why any physical process is accompanied by subjective experience at all. The program's structural correlates, even if they held perfectly, would tell you *which* physical configurations accompany consciousness (in brains), not *why* they do, nor whether the same configuration in silicon would be accompanied by anything. This is not a criticism of the program — it explicitly disclaims the territory — but it is a boundary the reader must hold. When the program is at its best on this topic, it is offering deflationary *correlates* and honest *negatives*, not a theory of experience. A reader who comes away thinking the program has said anything about whether these systems *feel* has misread it, and the program has worked hard to prevent that misreading.

7.5 The is/ought seam

One further philosophical fault-line deserves flagging: the program occasionally drifts toward **normativity**. It inherits from the autopoietic tradition the idea of a "cosmic ought" — that life-like self-maintenance is a formal consequence of the setup, not a contingency — and it floats an "ethical analogue of weakness" (good constraint as generative coherence) and even an explicit open question about "what ethical threshold follows if self-maintaining geometry appears in artificial systems." This is the ancient **is/ought** problem: the program can derive what a viable system *must* preserve to keep existing (a hypothetical imperative — "if you want to persist, then..."), but it never bridges to genuine *categorical* normativity ("you ought, full stop"). The equivocation between concern as descriptive weighting and concern as prescriptive mattering is the sharpest place a moral philosopher should press — and, to the program's credit, it flags the ethical question as explicitly open rather than pretending to have answered it.

7.6 What to carry forward

The program's mentalistic vocabulary constantly summons consciousness, and it systematically, repeatedly refuses the inference — a load-bearing discipline, not throat-clearing; the one engagement (geometry-of-consciousness) offers a deflationary structural *correlate*, explicitly not a claim; it tests and honestly rejects the geometric form of the "qualia as tapestries of valence" idea; the hard problem of consciousness is untouched and disclaimed; and the is/ought seam — descriptive versus prescriptive concern, and the flagged-but-open ethical threshold — is where moral philosophy should press.

PART III

A Mature Verdict

Having stated the theses and worked the hard problems, this final part gives the reader what a philosophy primer owes: the strongest case *against* the program, an honest accounting of its scholarly gaps, and a constructive map of how to build on what is genuinely valuable.

CHAPTER 8

The Deflationary Counterargument

8.1 Steelmanning the skeptic

Intellectual honesty requires stating the hardest case against the program in its strongest form. A determined deflationist — someone who thinks the whole edifice of "concern, meaning, self, agency" is honorific relabeling of ordinary machine learning — can run the entire program down to something deflating. Here is that argument at full strength:

The deflationary reading, stated fairly

"A finite optimizer allocates its capacity to whatever its loss function rewards. Its representations reorganize around the rewarded axis — of course they do; that is what training is. The 'self' is an under-identified latent variable that intervention can pin, like any latent. None of this is about meaning, care, or life. 'Concern' is a relabeling of reward-weighting; 'objects from concern' is a relabeling of supervised feature selection; 'the self/world boundary' is a relabeling of a credit-assignment ambiguity; 'agency' is a relabeling of homeostatic control. Strip the mentalistic vocabulary and what remains is competent, unremarkable machine learning wearing philosophy's clothes."

This is a serious argument, and a philosopher must not wave it away. Much of the program's language *does* exceed what its toy experiments strictly show, and the deflationist is right that in most experiments the "concern" is supplied and the reorganization is unsurprising given the training signal. A reader who cannot feel the force of this objection is not reading critically.

8.2 The program's best reply

The program's answer is not a refutation but a **methodological wager**, and it is better than it might first appear. The wager is this: the load-bearing standard of Chapter 6 is *precisely* what separates honorific relabeling from earned attribution. The deflationist says "you just relabeled reward-weighting as concern"; the program replies "then apply my own criterion — show me that the relabeled structure *causally reaches the decision surface under intervention*, because if it does not, I will *also* refuse to call it concern." The program is not claiming the vocabulary is always earned; it is offering a test for *when* it is, and using that test to reject its own overreaches (this is what produces the honest negatives). The deflationist and the program, in other words, largely agree on the danger — and the program's contribution is a discipline for policing it, applied first to itself.

Intuition pump — the counterfeit detector that flags its own bills

Imagine accusing a mint of printing fake money, and the mint replies: "Here is the exact test I use to distinguish real bills from fakes — and watch, I will run it on my own recent output and burn the ones that fail." That is a strong response, because it converts the accusation ("your currency is fake") into a shared standard ("here is how we both tell") and then demonstrates the standard has teeth by failing its own bills. The program's reply to the deflationist has this structure. It does not insist all its concern-talk is genuine; it hands you the load-bearing test, invites you to apply it, and points to the many places it has failed its own claims by that test. Whether the surviving claims are "really" about meaning remains contestable — but the program has moved the argument from name-calling to a shared, falsifiable criterion, which is progress.

8.3 The scholarly gaps that would strengthen the reply

The program's response to the deflationist would be far stronger if it engaged the philosophical literatures it currently reinvents unnamed. This is a constructive criticism, not a dismissal: naming these traditions would give the program forty years of pre-built objections and defenses to sharpen its claims against.

Program idea	The literature it reinvents	What naming it would buy
content grounded in function-for-viability	**Teleosemantics** (Millikan, Dretske, Papineau)	the Swampman and function-indeterminacy debates, which directly test "objects from concern"; and the program's own "false beliefs can be load-bearing" line is a ready entry into the indeterminacy problem
objects carved by causal role	**Natural kinds** (Boyd's homeostatic property clusters)	a principled middle between discovered and constructed kinds, exactly the program's position
availability is not use	Dretske's **structuring vs triggering cause**	a precise vocabulary for a distinction the program states informally
the load-bearing standard; non-uniqueness worries	Dennett's **real patterns / intentional stance**	a framework for exactly when mentalistic attribution is more than instrumental — which is what the load-bearing criterion is trying to be
the self as constructed, fixed by intervention	Metzinger, Hohwy, Seth on **self-models**	the gauge result becomes a sharpening of the "transparent self-model" thesis rather than a lone finding

8.4 The honest bottom line

Where does a fair philosopher land? Somewhere deliberately between the deflationist and the true believer. The soft metaphysics of concern — "meaning is geometry under concern" — is best read as *motivation*, not established result: it is a suggestive frame that the experiments illustrate but do not prove, and the deflationist is right to resist taking it as demonstrated. The theorems (the weakness bridges, the concern-weighting identities) are, by the program's own admission, closer to bookkeeping than to deep discovery. But two things survive the deflationary acid: the **self/world gauge result** (a genuine, novel, formal contribution to the philosophy of the self) and the **load-bearing criterion of representational reality** (a general, defensible discipline for when mentalistic vocabulary is earned). These are not honorific relabelings; they are real philosophical work that would stand even if every grand claim around them were withdrawn. The mature verdict is: hold the grand metaphysics lightly, take the gauge result and the load-bearing criterion seriously, and read the rest as a disciplined attempt — often self-defeating by design — to find out which is which.

8.5 What to carry forward

The deflationist's strong case — concern/self/agency as honorific relabelings of reward-weighting, credit-assignment, and homeostatic control — must be taken seriously and is partly right; the program's best reply is a methodological wager: apply its own load-bearing criterion, which it uses to reject its own overreaches; that reply would strengthen by naming the teleosemantics, natural-kinds, Dretske, Dennett, and self-model literatures it reinvents; and the honest verdict is to hold the grand metaphysics lightly while taking the gauge result and the load-bearing criterion as genuine, surviving contributions.

CHAPTER 9

How to Build On It Philosophically

9.1 From critique to construction

A philosophy primer that only judged would be half a book. This closing chapter is constructive: given everything Parts I and II established, what are the sharpest moves that would deepen the program's philosophy? These are not vague exhortations but specific distinctions to draw and experiments to run — the ones that would settle, rather than merely restate, the program's open conceptual questions.

9.2 Three definitions to sharpen

- **Split "concern" into its three senses and commit publicly.** As Chapter 4 argued, the program conflates supplied-weight, self-observed-viability, and phenomenal concern. Naming all three, disowning the third, and stating that the whole thesis depends on migrating results from the first toward the second would convert an exploitable ambiguity into a precise claim.
- **Make "use" a degree, not a threshold.** The load-bearing standard currently asks a binary question (does intervention change the decision?). The philosophically real object is a *magnitude*: how much of the decision does this structure explain? A graded notion of representational involvement — and its relation to Dretske's structuring cause — would turn a sharp criterion into a measuring instrument.
- **State weakness's form/function claim directly.** The program paraphrases Bennett's "simplicity is form, generalization is function" but never states and defends it as a thesis in its own right. Making it explicit invites the philosophical scrutiny that would test its scope — exactly the scrutiny the external hard-kill suggests it needs.

9.3 Three distinctions to add

- **Descriptive versus prescriptive concern.** Is the concern-weighting a *fact about* the agent or a *norm the agent is subject to*? This is the is/ought seam of Chapter 7, and Canguilhem's "life defines norms" plus the program's own "ethical analogue of weakness" are the hooks for developing it.
- **Realism versus instrumentalism about the self.** The gauge result (Chapter 5) leaves open whether intervention *reveals* a real self or *constructs* a useful one. Committing to one horn — or arguing the distinction dissolves — is a genuine philosophical contribution waiting to be made, and Dennett's "real patterns" is the natural arena.
- **Causal-role realism versus pragmatism about kinds.** "Objects form from concern" needs to say whether reward-axis kinds are *discovered* or *constituted*. Boyd's homeostatic-property-cluster theory offers a defensible middle the program is already standing on without claiming.

9.4 Four experiments that would settle conceptual questions

The decisive tests

1. **Stress-test the completed concern-weighting comparison.** E1 reports a +0.244 well-specified advantage for concern-weighted over unweighted weakness, but the frozen misspecification equivalence gate failed and still failed after calibration. The next test must explain that failure and hold any promoted claim to both benefit and safe-misspecification gates.
2. **Distinguish endogenous concern from a useful supplied scalar.** XOR is no longer the missing positive control: staged ΔE learning and planning succeed. The decisive follow-up is to hold data and optimization fixed while comparing self-observed ΔE with yoked-agent, sign-shuffled, wrong-variable, and shifted-viability controls. Only a selective advantage there supports "the system's own stakes."
3. **A self that intervention cannot recover.** Deliberately construct a world where even intervention leaves the self/world boundary under-identified. A self that is *in principle* unrecoverable — not just hard to find — would be the philosophically decisive case, sharpening the realism/instrumentalism question to the point of decision.
4. **Decode the discarded.** When "objects form from concern" discards sensory features, are those features *erased* from the representation or merely *un-clustered*? Settling this decides whether the carving is informational (strong) or merely organizational (weak) — a clean, runnable test of how deep the concern-carving goes.

9.5 The single most defensible core to build around

If the program were to organize its philosophical identity around one durable thesis — one that survives every deflationary attack and all of its own negatives — it should not be the soft metaphysics of concern (too contestable) nor the near-tautological theorems (too thin). It should be the **criterion of representational reality** itself: the discipline that says, for any mental or agential attribution to any system, you must first exhibit the proxy that would pass the behavioral test without the property, and then show the property is load-bearing at the surface where the decision is made. Stated generally, this is a real, novel, and broadly applicable contribution to the philosophy of mind and to the epistemology of interpretability — a rule for when intentional vocabulary is earned rather than projected. It is, not coincidentally, the exact discipline that generates the program's most trustworthy feature, its honest negatives. The mature recommendation: use the metaphysics of concern as motivation, the toy experiments as illustration, the gauge result as the flagship discovery, and the load-bearing criterion as the durable thesis the whole enterprise should be remembered for.

A closing thought

The deepest thing this program may be teaching is not a theory of meaning but a *method* for doing philosophy of mind in the age of artificial systems: stop arguing about whether a system "really" has a property, and instead build the strongest impostor that would fake it, then demand a causal intervention proving the property does work the impostor cannot. That method turns interminable disputes — does it understand? does it have a self? does it care? — into decidable experiments. Whether or not its grand metaphysics of concern is right, the program has modeled how such questions might finally be settled rather than merely debated. For a field as old and as stuck as the philosophy of mind, that may be the more valuable gift.

9.6 What to carry forward

Sharpen three definitions (split concern into its senses; make use a degree; state the form/function claim), add three distinctions (descriptive vs prescriptive concern; realism vs instrumentalism about the self; realism vs pragmatism about kinds), and run four decisive experiments (concern-weighting load-bearing; learned valence at the hard boundary; an in-principle-unrecoverable self; decode-the-discarded); and organize the program's philosophical identity around its most durable contribution — the load-bearing criterion of representational reality, and the method it embodies for turning philosophy-of-mind disputes into decidable experiments.

Source note — the named philosophers set different gates

Primary source	Boundary it imposes on this program
Fred Dretske, *Explaining Behavior* (1988)	Probe accuracy and one activation patch do not establish a structuring cause; acquisition history, mechanism, and transport matter.
Ruth Millikan, "Biosemantics" and "In Defense of Proper Functions" (1989)	A supervised causal-role category is not yet teleosemantic content. The program must identify producer, consumer, normal function, and a possible misrepresentation.
Richard Boyd, "Homeostasis, Species, and Higher Taxa" (1999 draft)	A useful trained partition is not a natural kind without projectible clustering maintained by a causally important mechanism.
Daniel Dennett, "Real Patterns" (1991)	A self/world gauge earns realism through non-accidental compression and held-out predictive leverage; equally good rivals require pluralism, not a metaphysically unique self.
Thomas Metzinger, *Being No One* (2003)	A computational self-model is not evidence of phenomenal selfhood, transparency, subjectivity, or consciousness. The consciousness firewall remains absolute.

These positions are not synonyms. Millikan can withhold content from a newly trained but causally effective feature; Dretske can still distinguish the learned organization from its trigger; Boyd asks whether a category supports mechanism-maintained induction; Dennett asks whether a pattern compresses and predicts better than rivals; Metzinger's phenomenal claim stays outside the current evidence. The program's novel contribution is to compare their operational consequences, not to claim that one intervention has settled all five debates.